

UTKARSH SINGH

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Professional Summary

Undergrad AI researcher specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning. I care about taste in engineering. From hackathon floors to research labs, I ship fast, break hard problems into working prototypes, and care deeply about systems that solve real user pain.

Education

Maharaja Agrasen Institute of Technology [GGSIPU] New Delhi, India
B.Tech in Computer Science Engineering, Specialization in Artificial Intelligence Aug. 2025 - May 2029
(9.3 CGPA)

Kamal Public Sr. Sec. School [CBSE] New Delhi, India
Class XII PCM with Computer Science (92.4%) April 2011 - May 2025

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C++, C

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, PyTorch, TensorFlow, Transformers, Hugging Face, scikit-learn

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs

Frontend: React, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, SQLite

MLOps & Cloud: Docker, Git, GitHub Actions (CI/CD), AWS (S3, EC2, Lambda), GCP, Vercel, Render, Railway

Libraries & Tools: NumPy, Pandas, LangChain, Matplotlib, OpenCV, Postman, Jupyter

Projects

KAVACH: Fraud Risk Manager for Indian D2C Merchants

- Python, FastAPI, XGBoost, SHAP, NetworkX, Next.js: built a calibrated return-risk scorer paired with graph-based fraud ring detection over shared phones, devices, VPAs and fuzzy-matched addresses.
- Defence-only risk manager for Indian D2C merchants targeting COD return abuse and coordinated chargebacks that erode 5-15% of gross margin.
- Ring detection scored 0.889 precision at 1.000 recall on held-out data; every flagged order ships with SHAP attributions and a SHA-256 hash-chained audit trail.

NETRA: AI Growth Copilot for Indian Kirana Stores

- Python, FastAPI, Llama 3.3 70B, Sarvam AI, React, Differential Privacy: built demand forecasting, non-collusive price benchmarking and credit scoring delivered via Soundbox audio, WhatsApp and web in 7 Indic languages.
- AI growth copilot for Indian Kirana stores serving 90% of a \$600B+ grocery market, enabling solo merchants to make data-backed pricing and inventory decisions.
- Enforces a five-layer privacy boundary using k-anonymity cohort gating and Laplace DP to block competitor surveillance and prevent price collusion.

ATLASS: Adaptive Taste Learning And Suggestion System

- JavaScript, SVD, LSA/TF-IDF, WebGL, GSAP, TMDb API: engineered SVD fold-in blended 70/30 with a 48-dimensional LSA content profile, pre-trained weights shipped as ~700KB JSON, re-scoring in under 50ms.
- Hybrid movie recommender running entirely in-browser with zero inference server, built on MovieLens (9,742 films, 100K+ ratings).
- WebGL gallery renders recommendation cards along a parametric arc with GSAP transitions for an interactive browsing experience.

Achievements

Top 10 Finalist, Paytm Build for India AI Hackathon 2026

- Finished Top 10 nationally; built NETRA, an AI growth copilot for Indian merchants with privacy-preserving market intelligence using k-anonymity and differential privacy.

Accepted Research Paper at ICML-Scale (International Conference on Machine Learning - Scale) 2026

- Co-authored **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding**, accepted at ICML-Scale 2026 as a second-year undergraduate.

Honoured, Award of Excellence 2025

- Honored with the Academic Excellence Award, presented by Shri Suresh Jain (National Organization Secretary, Bharat Vikas Parishad) and Shri Vijay Kumar (Joint Commissioner of Police).

Finalist, ET Gen AI Hackathon 2026

- Co-built an Autonomous Multi-Agent AI Code Intelligence & Remediation Platform; ranked Top Performing out of 3000+ candidates nationwide.